

Московский Авиационный Институт

(Национальный Исследовательский Университет)

Институт №8 “Компьютерные науки и прикладная математика”

Кафедра №806 “Вычислительная математика и программирование”

Лабораторная работа №5 по курсу

«Операционные системы»

Группа: М8О-209БВ-24

Студент: Буланов М.С.

Преподаватель: Миронов Е.С.

Оценка: _____

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Постановка задачи

Цель работы:

Приобретение практических навыков диагностики работы программного обеспечения.

Задание:

При выполнении лабораторных работ по курсу ОС необходимо продемонстрировать ключевые системные вызовы, которые в них используются и то, что их использование соответствует варианту ЛР.

Общий метод и алгоритм решения:

Strace - утилита командной строки для трассировки системных вызовов и сигналов в операционных системах Linux и других Unix-подобных системах. Её основное назначение - перехват и запись взаимодействия между пользовательским процессом и ядром операционной системы в реальном времени что полезно для отладки, анализа производительности и диагностики ошибок. Strace работает, используя механизм ядра ptrace (process trace), который позволяет одному процессу наблюдать и контролировать выполнение другого. При запуске с целевой программой strace перехватывает каждый системный вызов (например, открытие файла, запись в сеть, выделение памяти) на границе перехода из пользовательского пространства в пространство ядра и обратно. Это позволяет видеть не только факт вызова, но и его аргументы, возвращаемое значение и код ошибки (errno), если вызов завершился неудачно.

Основные флаги:

- p PID - присоединиться к уже работающему процессу с указанным идентификатором.
- c - подсчитать статистику по системным вызовам (время, вызовы, ошибки) и вывести сводку по завершении.
- f - трассировать также все дочерние процессы, созданные с помощью fork().
- e - фильтрация вывода
- o файл - вывести вывод трассировки в указанный файл вместо stderr.
- s размер - увеличить максимальную длину выводимых строк аргументов (по умолчанию часто 32 символа).
- t - выводить время в формате ЧЧ:ММ:СС при каждом вызове.
- T - показывать время, затраченное на каждый системный вызов.
- y - подробно выводить информацию о файловых дескрипторах (пути к файлам, сокетам).
- v - более подробный (verbose) вывод для некоторых вызовов.
- h - вывести справку по использованию.

Протокол работы

Lab1

```
root@bb4442151ed3:/workspaces/Operating_Systems/lab1/build# strace ./parent
execve("./parent", ["/parent"], 0x7ffe2097760 /* 27 vars */) = 0
brk(NULL)                               = 0xc46000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=33103, ...}) = 0
mmap(NULL, 33103, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fdbf5ff4000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0@\>\2\0\0\0\0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0755, st_size=1901536, ...}) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdbf5ff2000
mmap(NULL, 1914496, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fdbf5e1e000
mmap(0x7fdbf5e40000, 1413120, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x22000) = 0x7fdbf5e40000
mmap(0x7fdbf5f99000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x17b000) = 0x7fdbf5f99000
mmap(0x7fdbf5fe8000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c9000) = 0x7fdbf5fe8000
mmap(0x7fdbf5fee000, 13952, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fdbf5fee000
close(3)                                = 0
arch_prctl(ARCH_SET_FS, 0x7fdbf5ff3540) = 0
mprotect(0x7fdbf5fe8000, 16384, PROT_READ) = 0
mprotect(0x403000, 4096, PROT_READ)     = 0
mprotect(0x7fdbf6027000, 4096, PROT_READ) = 0
munmap(0x7fdbf5ff4000, 33103)           = 0
pipe([3, 4])                            = 0
pipe([5, 6])                            = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
brk(NULL)                               = 0xc46000
```

```

brk(0xc67000)                = 0xc67000

fstat(0, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0

write(1, "\320\222\320\262\320\265\320\264\320\270\321\202\320\265 \320\270\320\274\321\217
\321\204\320\260\320\271\320\273\320\260"..., 77Введите имя файла для дочернего процесса: ) = 77

read(0, ww

"ww\n", 1024)                = 3

clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD,
child_tidptr=0x7fdbf5ff3810) = 4182

close(3)                     = 0

close(6)                     = 0

write(1, "\320\224\320\276\321\207\320\265\321\200\320\275\320\270\320\271
\320\277\321\200\320\276\321\206\320\265\321\201\321\201 "..., 114Дочерний процесс создан.
Вводите команды (числа через пробел):

) = 114

write(1, "> ", 2> )          = 2

read(0, 8 4 2

"8 4 2\n", 1024)            = 6

write(4, "8 4 2", 5)         = 5

write(4, "\n", 1)            = 1

read(5, "\320\224\320\276\321\207\320\265\321\200\320\275\320\270\320\271
\320\277\321\200\320\276\321\206\320\265\321\201\321\201 "..., 1023) = 57

write(1, "\320\224\320\276\321\207\320\265\321\200\320\275\320\270\320\271
\320\277\321\200\320\276\321\206\320\265\321\201\321\201:"..., 90Дочерний процесс: Дочерний
процесс готов. Файл: ww

) = 90

wait4(4182, 0x7ffcba3eb3fc, WNOHANG, NULL) = 0

write(1, "> ", 2> )          = 2

read(0, 2 0

"2 0\n", 1024)              = 4

write(4, "2 0", 3)           = 3

write(4, "\n", 1)            = 1

read(5, "OK: Command processed successful"..., 1023) = 35

write(1, "\320\224\320\276\321\207\320\265\321\200\320\275\320\270\320\271
\320\277\321\200\320\276\321\206\320\265\321\201\321\201:"..., 68Дочерний процесс: ОК:
Command processed successfully

) = 68

```

```
wait4(4182, 0x7ffcba3eb3fc, WNOHANG, NULL) = 0
write(1, "> ", 2) = 2
read(0, 0xc466b0, 1024) = ? ERESTARTSYS (To be restarted if SA_RESTART is set)
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=4182, si_uid=0, si_status=0,
si_utime=0, si_stime=0} ---
read(0, 2
"2\n", 1024) = 2
write(4, "2", 1) = -1 EPIPE (Broken pipe)
--- SIGPIPE {si_signo=SIGPIPE, si_code=SI_USER, si_pid=4148, si_uid=0} ---
+++ killed by SIGPIPE +++
root@bb4442151ed3:/workspaces/Operating_Systems/lab1/build#
```

Lab2

```
root@c1cfa1e2c3:/workspaces/Operating_Systems/lab2/build# strace -f ./naive_search -t 4 "ll"
"allaalllalllalllalllalll"
execve("./naive_search", ["/naive_search", "-t", "4", "ll", "allaalllalllalllalllalll"], 0x7ffc69a9be18
/* 28 vars */) = 0
brk(NULL) = 0x565184552000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffd1676e280) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f2650c66000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f2650c62000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"..., 832) =
832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784,
64) = 784
pread64(3, "\4\0\0\0\0\0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48,
848) = 48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\00{\f\225\==\201\327\312\301P\32$\230\266\235"...,
68, 896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0
```

```

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784,
64) = 784

mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f2650a39000

mprotect(0x7f2650a61000, 2023424, PROT_NONE) = 0

mmap(0x7f2650a61000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f2650a61000

mmap(0x7f2650bf6000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0x1bd000) = 0x7f2650bf6000

mmap(0x7f2650c4f000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f2650c4f000

mmap(0x7f2650c55000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f2650c55000

close(3) = 0

mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f2650a36000

arch_prctl(ARCH_SET_FS, 0x7f2650a36740) = 0

set_tid_address(0x7f2650a36a10) = 26724

set_robust_list(0x7f2650a36a20, 24) = 0

rseq(0x7f2650a370e0, 0x20, 0, 0x53053053) = 0

mprotect(0x7f2650c4f000, 16384, PROT_READ) = 0

mprotect(0x565183e39000, 4096, PROT_READ) = 0

mprotect(0x7f2650ca0000, 8192, PROT_READ) = 0

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0

munmap(0x7f2650c62000, 12460) = 0

newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x3), ...}, AT_EMPTY_PATH) = 0

getrandom("\xdc\x5e\x25\xe7\xbb\xa5\x07\x01", 8, GRND_NONBLOCK) = 8

brk(NULL) = 0x565184552000

brk(0x565184573000) = 0x565184573000

write(1, "Number of threads: 4\n", 21Number of threads: 4
) = 21

write(1, "Starting search with 4 threads\n", 31Starting search with 4 threads
) = 31

write(1, "Text length: 29, Pattern: 'll' ("..., 41Text length: 29, Pattern: 'll' (2 chars)
) = 41

```

```
write(1, "Chunk size: 7 bytes\n", 20Chunk size: 7 bytes
```

```
) = 20
```

```
rt_sigaction(SIGRT_1, {sa_handler=0x7f2650aca870, sa_mask=[],  
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO, sa_restorer=0x7f2650a7b520},  
NULL, 8) = 0
```

```
rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
```

```
mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) =  
0x7f2650235000
```

```
mprotect(0x7f2650236000, 8388608, PROT_READ|PROT_WRITE) = 0
```

```
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
```

```
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SY  
SVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID,  
child_tid=0x7f2650a35910, parent_tid=0x7f2650a35910, exit_signal=0, stack=0x7f2650235000,  
stack_size=0x7fff00, tls=0x7f2650a35640}, 88) = -1 ENOSYS (Function not implemented)
```

```
clone(child_stack=0x7f2650a34ef0,  
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSE  
M|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTIDstrace: Process 26725  
attached
```

```
, parent_tid=[26725], tls=0x7f2650a35640, child_tidptr=0x7f2650a35910) = 26725
```

```
[pid 26725] rseq(0x7f2650a35fe0, 0x20, 0, 0x53053053 <unfinished ...>
```

```
[pid 26724] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
```

```
[pid 26725] <... rseq resumed> = 0
```

```
[pid 26724] <... rt_sigprocmask resumed>NULL, 8) = 0
```

```
[pid 26725] set_robust_list(0x7f2650a35920, 24 <unfinished ...>
```

```
[pid 26724] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,  
-1, 0 <unfinished ...>
```

```
[pid 26725] <... set_robust_list resumed> = 0
```

```
[pid 26724] <... mmap resumed> = 0x7f264fa34000
```

```
[pid 26725] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
```

```
[pid 26724] mprotect(0x7f264fa35000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>
```

```
[pid 26725] <... rt_sigprocmask resumed>NULL, 8) = 0
```

```
[pid 26724] <... mprotect resumed> = 0
```

```
[pid 26725] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>
```

```
[pid 26724] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>
```

```
[pid 26725] <... rt_sigprocmask resumed>NULL, 8) = 0
```

```
[pid 26724] <... rt_sigprocmask resumed>[], 8) = 0
```

[pid 26725] madvise(0x7f2650235000, 8368128, MADV_DONTNEED <unfinished ...>

[pid 26724]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SY
SVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID,
child_tid=0x7f2650234910, parent_tid=0x7f2650234910, exit_signal=0, stack=0x7f264fa34000,
stack_size=0x7fff00, tls=0x7f2650234640} <unfinished ...>

[pid 26725] <... madvise resumed> = 0

[pid 26724] <... clone3 resumed>, 88) = -1 ENOSYS (Function not implemented)

[pid 26725] exit(0 <unfinished ...>

**[pid 26724] clone(child_stack=0x7f2650233ef0,
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSE
M|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID <unfinished ...>**

[pid 26725] <... exit resumed> = ?

strace: Process 26726 attached

[pid 26725] +++ exited with 0 +++

[pid 26724] <... clone resumed>, parent_tid=[26726], tls=0x7f2650234640,
child_tidptr=0x7f2650234910) = 26726

[pid 26726] rseq(0x7f2650234fe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 26724] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 26726] <... rseq resumed> = 0

[pid 26724] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26726] set_robust_list(0x7f2650234920, 24 <unfinished ...>

[pid 26724] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,
-1, 0 <unfinished ...>

[pid 26726] <... set_robust_list resumed>) = 0

[pid 26724] <... mmap resumed> = 0x7f264f233000

[pid 26726] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 26724] mprotect(0x7f264f234000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>

[pid 26726] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26724] <... mprotect resumed> = 0

[pid 26726] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>

[pid 26724] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>

[pid 26726] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26724] <... rt_sigprocmask resumed>[], 8) = 0

[pid 26726] madvise(0x7f264fa34000, 8368128, MADV_DONTNEED <unfinished ...>

[pid 26724]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SY
SVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID,
child_tid=0x7f264fa33910, parent_tid=0x7f264fa33910, exit_signal=0, stack=0x7f264f233000,
stack_size=0x7fff00, tls=0x7f264fa33640} <unfinished ...>

[pid 26726] <... madvise resumed> = 0

[pid 26724] <... clone3 resumed>, 88) = -1 ENOSYS (Function not implemented)

[pid 26726] exit(0 <unfinished ...>

**[pid 26724] clone(child_stack=0x7f264fa32ef0,
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSE
M|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID <unfinished ...>**

[pid 26726] <... exit resumed> = ?

[pid 26726] +++ exited with 0 +++

strace: Process 26727 attached

**[pid 26724] <... clone resumed>, parent_tid=[26727], tls=0x7f264fa33640,
child_tidptr=0x7f264fa33910) = 26727**

[pid 26727] rseq(0x7f264fa33fe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 26724] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 26727] <... rseq resumed> = 0

[pid 26724] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26727] set_robust_list(0x7f264fa33920, 24 <unfinished ...>

[pid 26724] mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,
-1, 0 <unfinished ...>

[pid 26727] <... set_robust_list resumed> = 0

[pid 26724] <... mmap resumed> = 0x7f264ea32000

[pid 26727] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 26724] mprotect(0x7f264ea33000, 8388608, PROT_READ|PROT_WRITE <unfinished ...>

[pid 26727] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26724] <... mprotect resumed> = 0

[pid 26727] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>

[pid 26724] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>

[pid 26727] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26724] <... rt_sigprocmask resumed>[], 8) = 0

[pid 26727] madvise(0x7f264f233000, 8368128, MADV_DONTNEED <unfinished ...>

[pid 26724]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SY
SVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID,
child_tid=0x7f264f232910, parent_tid=0x7f264f232910, exit_signal=0, stack=0x7f264ea32000,
stack_size=0x7fff00, tls=0x7f264f232640} <unfinished ...>

[pid 26727] <... madwrite resumed> = 0

[pid 26724] <... clone3 resumed>, 88) = -1 ENOSYS (Function not implemented)

[pid 26727] exit(0 <unfinished ...>

[pid 26724] clone(child_stack=0x7f264f231ef0,
flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSE
M|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID <unfinished ...>

[pid 26727] <... exit resumed> = ?

strace: Process 26728 attached

[pid 26727] +++ exited with 0 +++

[pid 26724] <... clone resumed>, parent_tid=[26728], tls=0x7f264f232640,
child_tidptr=0x7f264f232910) = 26728

[pid 26728] rseq(0x7f264f232fe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 26724] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 26728] <... rseq resumed> = 0

[pid 26724] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 26728] set_robust_list(0x7f264f232920, 24 <unfinished ...>

[pid 26724] futex(0x7f264f232910, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 26728,
NULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 26728] <... set_robust_list resumed>) = 0

[pid 26728] rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

[pid 26728] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 26728] madwrite(0x7f264ea32000, 8368128, MADV_DONTNEED) = 0

[pid 26728] exit(0) = ?

[pid 26724] <... futex resumed> = 0

[pid 26728] +++ exited with 0 +++

write(1, "\n", 1

) = 1

write(1, "Search completed. Found 16 occur"..., 40Search completed. Found 16 occurrences:

) = 40

write(1, "Position 1\n", 11Position 1

```
)      = 11
write(1, "Position 5\n", 11Position 5
)      = 11
write(1, "Position 6\n", 11Position 6
)      = 11
write(1, "Position 9\n", 11Position 9
)      = 11
write(1, "Position 10\n", 12Position 10
)      = 12
write(1, "Position 11\n", 12Position 11
)      = 12
write(1, "Position 14\n", 12Position 14
)      = 12
write(1, "Position 15\n", 12Position 15
)      = 12
write(1, "Position 16\n", 12Position 16
)      = 12
write(1, "Position 17\n", 12Position 17
)      = 12
write(1, "Position 18\n", 12Position 18
)      = 12
write(1, "Position 21\n", 12Position 21
)      = 12
write(1, "Position 22\n", 12Position 22
)      = 12
write(1, "Position 23\n", 12Position 23
)      = 12
write(1, "Position 26\n", 12Position 26
)      = 12
write(1, "Position 27\n", 12Position 27
)      = 12
write(1, "\n", 1
```

) = 1

write(1, "Execution time: 0.013274 seconds"..., 33Execution time: 0.013274 seconds

) = 33

exit_group(0) = ?

+++ exited with 0 +++

Lab_3:

```
execve("./parent", ["./parent"], 0x7ffd4e344c68 /* 28 vars */) = 0
brk(NULL) = 0x55c4a863f000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffeb2a9d6e0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f822b344000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f822b340000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0"... , 48, 848) =
48
pread64(3, "\4\0\0\0\24\0\0\0\3\0\0\0GNU\00{\f\225\=\201\327\312\301P\32$\230\266\235"...
, 68, 896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f822b117000
mprotect(0x7f822b13f000, 2023424, PROT_NONE) = 0
mmap(0x7f822b13f000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0x28000) = 0x7f822b13f000
mmap(0x7f822b2d4000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000)
= 0x7f822b2d4000
mmap(0x7f822b32d000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3,
0x215000) = 0x7f822b32d000
mmap(0x7f822b333000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -
1, 0) = 0x7f822b333000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f822b114000
arch_prctl(ARCH_SET_FS, 0x7f822b114740) = 0
set_tid_address(0x7f822b114a10) = 1747
set_robust_list(0x7f822b114a20, 24) = 0
rseq(0x7f822b1150e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f822b32d000, 16384, PROT_READ) = 0
mprotect(0x55c4a7452000, 4096, PROT_READ) = 0
mprotect(0x7f822b37e000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f822b340000, 12460) = 0
[SIGNALS] rt_sigaction(SIGUSR1, {sa_handler=0x55c4a7450509, sa_mask=[], sa_flags=SA_RESTOR
ER|SA_RESTART, sa_restorer=0x7f822b159520}, NULL, 8) = 0
[SHARED FILES] openat(AT_FDCWD, "cmd.bin", O_RDWR|O_CREAT|O_TRUNC, 0600) = 3
[SHARED FILES] ftruncate(3, 1096) = 0
[SHARED FILES] mmap(NULL, 1096, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f822b37d000
[SHARED FILES] close(3) = 0
[SHARED FILES] openat(AT_FDCWD, "resp.bin", O_RDWR|O_CREAT|O_TRUNC, 0600) = 3
[SHARED FILES] ftruncate(3, 1028) = 0
```

```

[SHARED FILES] mmap(NULL, 1028, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7f822b343000
[SHARED FILES] close(3) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY_PATH) =
0
getrandom("\xf7\xfa\xbb\x45\xd1\x79\xa4\x1c", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x55c4a863f000
brk(0x55c4a8660000) = 0x55c4a8660000
write(1, "\320\244\320\260\320\271\320\273 \320\264\320\273\321\217 \321\200\320\265\320\2
67\321\203\320\273\321\214\321\202\320\260"... , 40Файл для результатов: ) = 40
newfstatat(0, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY_PATH) =
0
read(0, ans
"ans \n", 1024) = 5
[CHILD PROC] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD
strace: Process 1785 attached
, child_tidptr=0x7f822b114a10) = 1785
[pid 1785] set_robust_list(0x7f822b114a20, 24 <unfinished ...>
[SEM/FUTEX] [pid 1747] futex(0x7f822b37d428, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, N
ULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>
[pid 1785] <... set_robust_list resumed>) = 0
[CHILD PROC] [pid 1785] execve("./child", ["child"], 0x7ffeb2a9d8b8 /* 28 vars */) = 0
[pid 1785] brk(NULL) = 0x564aed05f000
[pid 1785] arch_prctl(0x3001 /* ARCH_??? */, 0x7ffe7c530850) = -
1 EINVAL (Invalid argument)
[pid 1785] mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -
1, 0) = 0x7fcfd2933000
[pid 1785] access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
[pid 1785] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[pid 1785] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) =
0
[pid 1785] mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fcfd292f000
[pid 1785] close(3) = 0
[pid 1785] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[pid 1785] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"... ,
832) = 832
[pid 1785] pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0"... ,
784, 64) = 784
[pid 1785] pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0"... ,
48, 848) = 48
[pid 1785] pread64(3, "\4\0\0\0\24\0\0\0\3\0\0\0GNU\00{\f\225\\=\201\327\312\301P\32$\230
\266\235"... , 68, 896) = 68
[pid 1785] newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH)
= 0
[pid 1785] pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0"... ,
784, 64) = 784
[pid 1785] mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fcfd27060
00
[pid 1785] mprotect(0x7fcfd272e000, 2023424, PROT_NONE) = 0
[pid 1785] mmap(0x7fcfd272e000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_D
ENYWRITE, 3, 0x28000) = 0x7fcfd272e000
[pid 1785] mmap(0x7fcfd28c3000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3
, 0x1bd000) = 0x7fcfd28c3000
[pid 1785] mmap(0x7fcfd291c000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DE
NYWRITE, 3, 0x215000) = 0x7fcfd291c000
[pid 1785] mmap(0x7fcfd2922000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_AN
ONYMOUS, -1, 0) = 0x7fcfd2922000
[pid 1785] close(3) = 0
[pid 1785] mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -
1, 0) = 0x7fcfd2703000
[pid 1785] arch_prctl(ARCH_SET_FS, 0x7fcfd2703740) = 0
[pid 1785] set_tid_address(0x7fcfd2703a10) = 1785
[pid 1785] set_robust_list(0x7fcfd2703a20, 24) = 0

```

```

[pid 1785] rseq(0x7fcfd27040e0, 0x20, 0, 0x53053053) = 0
[pid 1785] mprotect(0x7fcfd291c000, 16384, PROT_READ) = 0
[pid 1785] mprotect(0x564aeba9d000, 4096, PROT_READ) = 0
[pid 1785] mprotect(0x7fcfd296d000, 8192, PROT_READ) = 0
[pid 1785] prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY
}) = 0
[pid 1785] munmap(0x7fcfd292f000, 12460) = 0
[SIGNALS] [pid 1785] rt_sigaction(SIGUSR1, {sa_handler=0x564aeba9b489, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7fcfd2748520}, NULL, 8) = 0
[SHARED FILES] [pid 1785] openat(AT_FDCWD, "cmd.bin", O_RDWR) = 3
[SHARED FILES] [pid 1785] mmap(NULL, 1096, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7fcfd296c000
[SHARED FILES] [pid 1785] close(3) = 0
[SHARED FILES] [pid 1785] openat(AT_FDCWD, "resp.bin", O_RDWR) = 3
[SHARED FILES] [pid 1785] mmap(NULL, 1028, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7fcfd2932000
[SHARED FILES] [pid 1785] close(3) = 0
[pid 1785] getrandom("\x06\xcf\xfa\x5e\x9e\x02\x96\x57", 8, GRND_NONBLOCK) = 8
[pid 1785] brk(NULL) = 0x564aed05f000
[pid 1785] brk(0x564aed080000) = 0x564aed080000
[OUTPUT FILE] [pid 1785] openat(AT_FDCWD, "ans ", O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
[pid 1785] getpid() = 1785
[pid 1785] newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=0, ...}, AT_EMPTY_PATH) = 0
[OUTPUT FILE] [pid 1785] write(3, "\320\224\320\276\321\207\320\265\321\200\320\275\320\270\320\271 \320\277\321\200\320\276\321\206\320\265\321\201\321\201 "..., 43) = 43
[SEM/FUTEX] [pid 1785] futex(0x7fcfd296c428, FUTEX_WAKE, 1 <unfinished ...>
[SEM/FUTEX] [pid 1747] <... futex resumed> = 0
[SEM/FUTEX] [pid 1785] <... futex resumed> = 1
[pid 1747] write(1, "Child says: READY\n", 18 <unfinished ...>
Child says: READY
[pid 1785] getppid( <unfinished ...>
[pid 1747] <... write resumed> = 18
[pid 1785] <... getppid resumed> = 1747
[pid 1747] write(1, "\320\224\320\276\321\207\320\265\321\200\320\275\320\270\320\271 PID : 1785\n", 27 <unfinished ...>
Дочерний PID: 1785
[SIGNALS] [pid 1785] kill(1747, SIGUSR1 <unfinished ...>
[pid 1747] <... write resumed> = 27
[SIGNALS] [pid 1785] <... kill resumed> = 0
[SIGNALS] [pid 1747] --
- SIGUSR1 {si_signo=SIGUSR1, si_code=SI_USER, si_pid=1785, si_uid=0} ---
[pid 1785] futex(0x7fcfd296c408, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>
[SIGNALS] [pid 1747] rt_sigreturn({mask=[]}) = 27
[pid 1747] write(1, "\320\222\320\262\320\276\320\264\320\270\321\202\320\265 \321\207\320\270\321\201\320\273\320\260 \321\207\320\265\321\200"..., 80Вводите числа через пробел ( 'exit' для выхода):
) = 80
[pid 1747] write(1, "> ", 2) = 2
[pid 1747] read(0, 16 2 2
"16 2 2\n", 1024) = 7
[SEM/FUTEX] [pid 1747] futex(0x7f822b37d408, FUTEX_WAKE, 1) = 1
[SEM/FUTEX] [pid 1785] <... futex resumed> = 0
[SIGNALS] [pid 1747] kill(1785, SIGUSR1 <unfinished ...>
[OUTPUT FILE] [pid 1785] write(3, "\320\232\320\276\320\274\320\260\320\275\320\264\320\260: 16 2 2\n\320\222\321\213\321\207\320\270\321"..., 93 <unfinished ...>
[SIGNALS] [pid 1747] <... kill resumed> = 0
[SEM/FUTEX] [pid 1747] futex(0x7f822b37d428, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>
[OUTPUT FILE] [pid 1785] <... write resumed> = 93
[SIGNALS] [pid 1785] --
- SIGUSR1 {si_signo=SIGUSR1, si_code=SI_USER, si_pid=1747, si_uid=0} ---

```

```

[ SIGNALS ] [ pid 1785 ] rt_sigreturn({mask=[]}) = 93
[ SEM/FUTEX ] [ pid 1785 ] futex(0x7fcfd296c428, FUTEX_WAKE, 1 <unfinished ...>
[ SEM/FUTEX ] [ pid 1747 ] <... futex resumed> = 0
[ SEM/FUTEX ] [ pid 1785 ] <... futex resumed> = 1
[ pid 1785 ] getppid( <unfinished ...>
[ pid 1747 ] write(1, "\320\236\321\202\320\262\320\265\321\202: OK\n", 15 <unfinished ...>
[ pid 1785 ] <... getppid resumed> = 1747
Ответ: OK
[ pid 1747 ] <... write resumed> = 15
[ pid 1747 ] write(1, "> ", 2 <unfinished ...>
> [ pid 1785 ] kill(1747, SIGUSR1 <unfinished ...>
[ pid 1747 ] <... write resumed> = 2
[ pid 1785 ] <... kill resumed> = 0
[ SIGNALS ] [ pid 1747 ] --
- SIGUSR1 {si_signo=SIGUSR1, si_code=SI_USER, si_pid=1785, si_uid=0} ---
[ pid 1785 ] futex(0x7fcfd296c408, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, NULL, FUTEX_B
ITSET_MATCH_ANY <unfinished ...>
[ SIGNALS ] [ pid 1747 ] rt_sigreturn({mask=[]}) = 2
[ pid 1747 ] read(0, 2 2 0
"2 2 0\n", 1024) = 6
[ SEM/FUTEX ] [ pid 1747 ] futex(0x7f822b37d408, FUTEX_WAKE, 1) = 1
[ SEM/FUTEX ] [ pid 1785 ] <... futex resumed> = 0
[ SIGNALS ] [ pid 1747 ] kill(1785, SIGUSR1 <unfinished ...>
[ OUTPUT FILE ] [ pid 1785 ] write(3, "\320\232\320\276\320\274\320\260\320\275\320\264\320\2
60: 2 2 0\n\320\222\321\213\321\207\320\270\321\201"... , 94 <unfinished ...>
[ SIGNALS ] [ pid 1747 ] <... kill resumed> = 0
[ SEM/FUTEX ] [ pid 1747 ] futex(0x7f822b37d428, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 0, N
ULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>
[ OUTPUT FILE ] [ pid 1785 ] <... write resumed> = 94
[ SIGNALS ] [ pid 1785 ] --
- SIGUSR1 {si_signo=SIGUSR1, si_code=SI_USER, si_pid=1747, si_uid=0} ---
[ SIGNALS ] [ pid 1785 ] rt_sigreturn({mask=[]}) = 94
[ SEM/FUTEX ] [ pid 1785 ] futex(0x7fcfd296c428, FUTEX_WAKE, 1) = 1
[ SEM/FUTEX ] [ pid 1747 ] <... futex resumed> = 0
[ pid 1747 ] write(1, "\320\236\321\202\320\262\320\265\321\202: CRITICAL: \320\264\320\265
\320\273\320\265\320\275"... , 51 <unfinished ...>
Ответ: CRITICAL: деление на ноль
[ pid 1785 ] getppid( <unfinished ...>
[ pid 1747 ] <... write resumed> = 51
[ pid 1785 ] <... getppid resumed> = 1747
[ pid 1747 ] write(1, "\n", 1
) = 1
[ SIGNALS ] [ pid 1785 ] kill(1747, SIGUSR1 <unfinished ...>
[ pid 1747 ] write(1, "!!! \320\236\320\250\320\230\320\221\320\232\320\220: \320\224\320\2
25\320\233\320\225\320\235\320\230\320\225"... , 51 <unfinished ...>
[ SIGNALS ] [ pid 1785 ] <... kill resumed> = 0
[ pid 1747 ] <... write resumed> = ? ERESTARTSYS (To be restarted if SA_RESTART is
set)
[ OUTPUT FILE ] [ pid 1785 ] close(3 <unfinished ...>
[ SIGNALS ] [ pid 1747 ] --
- SIGUSR1 {si_signo=SIGUSR1, si_code=SI_USER, si_pid=1785, si_uid=0} ---
[ SIGNALS ] [ pid 1747 ] rt_sigreturn({mask=[]}) = 1
[ pid 1747 ] write(1, "!!! \320\236\320\250\320\230\320\221\320\232\320\220: \320\224\320\2
25\320\233\320\225\320\235\320\230\320\225"... , 51!!! ОШИБКА: ДЕЛЕНИЕ НА НОЛЬ !!!
) = 51
[ OUTPUT FILE ] [ pid 1785 ] <... close resumed> = 0
[ pid 1747 ] write(1, "\320\227\320\260\320\262\320\265\321\200\321\210\320\265\320\275\320
\270\320\265 \321\200\320\260\320\261\320\276\321\202\321"... , 37 <unfinished ...>
Завершение работы...
[ SHARED FILES ] [ pid 1785 ] munmap(0x7fcfd296c000, 1096 <unfinished ...>
[ pid 1747 ] <... write resumed> = 37
[ SIGNALS ] [ pid 1747 ] kill(1785, SIGTERM) = 0

```

```

[ SIGNALS ] [ pid 1747 ] wait4(1785, <unfinished ...>
[ SHARED FILES ] [ pid 1785 ] <... munmap resumed> ) = 0
[ SIGNALS ] [ pid 1785 ] --
- SIGTERM { si_signo=SIGTERM, si_code=SI_USER, si_pid=1747, si_uid=0 } ---
[ SIGNALS ] [ pid 1785 ] +++ killed by SIGTERM +++
[ CHILD PROC ] <... wait4 resumed> NULL, 0, NULL ) = 1785
[ SIGNALS ] --
- SIGCHLD { si_signo=SIGCHLD, si_code=CLD_KILLED, si_pid=1785, si_uid=0, si_status=SIGTERM,
  si_utime=0, si_stime=0 } ---
[ SHARED FILES ] munmap(0x7f822b37d000, 1096) = 0
[ SHARED FILES ] munmap(0x7f822b343000, 1028) = 0
[ SHARED FILES ] unlink("cmd.bin") = 0
[ SHARED FILES ] unlink("resp.bin") = 0
write(1, "\320\240\320\276\320\264\320\270\321\202\320\265\320\273\321\214\321\201\320\272
\320\270\320\271 \320\277\321\200\320\276\321"... , 58Родительский процесс завершен.
) = 58
exit_group(0) = ?
+++ exited with 0 +++

```

Lab_4:

Статическая версия (compile_time_prog):

```

root@c1caffa1e2c3:/workspaces/Operating_Systems/lab4/build/bin# strace -
t ./compile_time_prog
12:13:19 execve("./compile_time_prog", [". /compile_time_prog"], 0x7ffffd84ee58 /* 28 vars
*/) = 0
12:13:19 brk(NULL) = 0x55ef91661000
12:13:19 arch_prctl(0x3001 /* ARCH_??? */, 0x7fff81d28ae0) = -1 EINVAL (Invalid argument)
12:13:19 mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -
1, 0) = 0x7fdb5fdb1000
12:13:19 access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/glibc-hwcap
s/x86-64-v3/libimpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/glibc-
hwcaps/x86-64-v3", 0x7fff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/glibc-hwcap
s/x86-64-v2/libimpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/glibc-
hwcaps/x86-64-v2", 0x7fff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/haswell/x86_64
/libimpl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/haswell/x8
6_64", 0x7fff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/haswell/libimp
l1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/haswell",
0x7fff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/x86_64/libimpl
1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/x86_64", 0
x7fff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls/libimpl1.so",
O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/tls", 0x7fff81
d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/haswell/x86_64/lib
impl1.so", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/haswell/x86_64
", 0x7fff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/haswell/libimpl1.s
o", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)

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12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/haswell", 0x7f
ff81d27d00, 0) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/x86_64/libimpl1.so
", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 newfstatat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/x86_64", 0x7ff
f81d27d00, 0) = -1 ENOENT (No such file or directory)
Подгруз моей библиотеки и отображение в память процесса
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/libimpl1.so", O_RD
ONLY|O_CLOEXEC) = 3
12:13:19 read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0"..., 832)
= 832
12:13:19 newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=15760, ...}, AT_EMPTY_PATH) = 0
12:13:19 mmap(NULL, 16448, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fdb5fdac000
12:13:19 mmap(0x7fdb5fdad000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x1000) = 0x7fdb5fdad000
12:13:19 mmap(0x7fdb5fdae000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2
000) = 0x7fdb5fdae000
12:13:19 mmap(0x7fdb5daf000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x2000) = 0x7fdb5daf000
12:13:19 close(3) = 0
12:13:19 openat(AT_FDCWD, "/workspaces/Operating_Systems/lab4/build/bin/libc.so.6", O_RDON
LY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
12:13:19 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
12:13:19 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
12:13:19 mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fdb5fda8000
12:13:19 close(3) = 0
12:13:19 openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
12:13:19 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"..., 832
) = 832
12:13:19 pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 78
4, 64) = 784
12:13:19 pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0"..., 4
8, 848) = 48
12:13:19 pread64(3, "\4\0\0\0\24\0\0\0\3\0\0\0GNU\00{\f\225\\=\201\327\312\301P\32$\230\26
6\235"..., 68, 896) = 68
12:13:19 newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
12:13:19 pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 78
4, 64) = 784
12:13:19 mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fdb5fb7f000
12:13:19 mprotect(0x7fdb5fba7000, 2023424, PROT_NONE) = 0
12:13:19 mmap(0x7fdb5fba7000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENY
WRITE, 3, 0x28000) = 0x7fdb5fba7000
12:13:19 mmap(0x7fdb5fd3c000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x1bd000) = 0x7fdb5fd3c000
12:13:19 mmap(0x7fdb5fd95000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0x215000) = 0x7fdb5fd95000
12:13:19 mmap(0x7fdb5fd9b000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONY
MOUS, -1, 0) = 0x7fdb5fd9b000
12:13:19 close(3) = 0
12:13:19 openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
12:13:19 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0"..., 832)
= 832
12:13:19 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) = 0
12:13:19 mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fdb5fa98000
12:13:19 mmap(0x7fdb5faa6000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0xe000) = 0x7fdb5faa6000
12:13:19 mmap(0x7fdb5fb22000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x8a000) = 0x7fdb5fb22000
12:13:19 mmap(0x7fdb5fb7d000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0xe4000) = 0x7fdb5fb7d000
12:13:19 close(3) = 0
```

```

12:13:19 mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -
1, 0) = 0x7fdb5fa95000
12:13:19 arch_prctl(ARCH_SET_FS, 0x7fdb5fa95740) = 0
12:13:19 set_tid_address(0x7fdb5fa95a10) = 8241
12:13:19 set_robust_list(0x7fdb5fa95a20, 24) = 0
12:13:19 rseq(0x7fdb5fa960e0, 0x20, 0, 0x53053053) = 0
12:13:19 mprotect(0x7fdb5fd95000, 16384, PROT_READ) = 0
12:13:19 mprotect(0x7fdb5fb7d000, 4096, PROT_READ) = 0
12:13:19 mprotect(0x7fdb5fdaf000, 4096, PROT_READ) = 0
12:13:19 mprotect(0x55ef912c7000, 4096, PROT_READ) = 0
12:13:19 mprotect(0x7fdb5fdeb000, 8192, PROT_READ) = 0
12:13:19 prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY})
= 0
12:13:19 munmap(0x7fdb5fda8000, 12460) = 0
12:13:19 newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY
_PATH) = 0
12:13:19 getrandom("\x1c\xf5\x00\x75\xae\x6f\x36\xb2", 8, GRND_NONBLOCK) = 8
12:13:19 brk(NULL) = 0x55ef91661000
12:13:19 brk(0x55ef91682000) = 0x55ef91682000
12:13:19 write(1, "=== \320\237\321\200\320\276\320\263\321\200\320\260\320\274\320\274\32
0\260 \321\201 \321\201\320\272\320\276"... , 90=== Программа с скомпилированными библиотек
ами ===
) = 90
12:13:19 write(1, "\320\232\320\276\320\274\320\260\320\275\320\264\321\213:\n", 16Команды
:
) = 16
12:13:19 write(1, " 0 - \320\262\321\213\321\205\320\276\320\264\n", 17 0 - выход
) = 17
12:13:19 write(1, " 1 \321\207\320\270\321\201\320\273\320\276 - \320\277\320\265\321\200
\320\265\320\262\320\276\320\264 "... , 43 1 число - перевод числа
) = 43
12:13:19 write(1, " 2 A B e - \320\270\320\275\321\202\320\265\320\263\321\200\320\260\32
0\273 sin"... , 36 2 A B e - интеграл sin(x)
) = 36
12:13:19 write(1, "\n", 1
) = 1
12:13:19 newfstatat(0, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY
_PATH) = 0
12:13:19 write(1, "> ", 2> ) = 2
12:13:19 read(0, 1
"1\n", 1024) = 2
12:13:24 write(1, "\320\236\321\210\320\270\320\261\320\272\320\260: \320\262\320\262\320\
265\320\264\320\270\321\202\320\265 \321\207\320"... , 68Ошибка: введите число после команд
ы 1
) = 68
12:13:24 write(1, "> ", 2> ) = 2
12:13:24 read(0, 7
"7\n", 1024) = 2
12:13:26 write(1, "\320\235\320\265\320\270\320\267\320\262\320\265\321\201\321\202\320\27
5\320\260\321\217 \320\272\320\276\320\274\320\260\320"... , 41Неизвестная команда: 7
) = 41
12:13:26 write(1, "> ", 2> ) = 2
12:13:26 read(0, 1 7
"1 7 \n", 1024) = 5
12:13:40 write(1, "\320\240\320\265\320\267\321\203\320\273\321\214\321\202\320\260\321\20
2: 111\n", 24Результат: 111
) = 24
12:13:40 write(1, "> ", 2> ) = 2
12:13:40 read(0, 2 -1 1 0.1
"2 -1 1 0.1\n", 1024) = 11
12:13:50 write(1, "[\320\221\320\270\320\261\320\273\320\270\320\276\321\202\320\265\320\2
72\320\260 1] \320\234\320\265\321\202\320"... , 67[Библиотека 1] Метод ПРЯМОУГОЛЬНИКОВ

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```

) = 67
12:13:50 write(1, "\\320\\240\\320\\265\\320\\267\\321\\203\\320\\273\\321\\214\\321\\202\\320\\260\\321\\20
2 \\320\\270\\320\\275\\321\\202\\320\\265\\320\\263\\321\\200\\320"... , 48Результат интеграла: 0.00000
0
) = 48
12:13:50 write(1, "> ", 2> ) = 2
12:13:50 read(0, 0
"0\\n", 1024) = 2
12:13:54 write(1, "\\320\\222\\321\\213\\321\\205\\320\\276\\320\\264\\n", 11Выход
) = 11
12:13:54 exit_group(0) = ?
12:13:54 +++ exited with 0 +++

```

Динамическая загрузка (runtime_prog)

```

root@c1caffa1e2c3:/workspaces/Operating_Systems/lab4/build/bin# strace -t ./runtime_prog
12:15:52 execve("./runtime_prog", [ "./runtime_prog" ], 0x7ffec97884d8 /* 28 vars */) = 0
12:15:52 brk(NULL) = 0x55df156db000
12:15:52 arch_prctl(0x3001 /* ARCH_??? */, 0x7ffed5c01230) = -1 EINVAL (Invalid argument)
12:15:52 mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -
1, 0) = 0x7f7517cf6000
12:15:52 access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
12:15:52 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
12:15:52 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
12:15:52 mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f7517cf2000
12:15:52 close(3) = 0
12:15:52 openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
12:15:52 read(3, "\\177ELF\\2\\1\\1\\3\\0\\0\\0\\0\\0\\0\\0\\0\\3\\0>\\0\\1\\0\\0\\0P\\237\\2\\0\\0\\0\\0\\0"... , 832
) = 832
12:15:52 pread64(3, "\\6\\0\\0\\0\\4\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0"... , 78
4, 64) = 784
12:15:52 pread64(3, "\\4\\0\\0\\0 \\0\\0\\0\\5\\0\\0\\0GNU\\0\\2\\0\\0\\300\\4\\0\\0\\0\\3\\0\\0\\0\\0\\0\\0\\0"... , 4
8, 848) = 48
12:15:52 pread64(3, "\\4\\0\\0\\0\\24\\0\\0\\0\\3\\0\\0\\0GNU\\00{\\f\\225\\|=\\201\\327\\312\\301P\\32$\\230\\26
6\\235"... , 68, 896) = 68
12:15:52 newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
12:15:52 pread64(3, "\\6\\0\\0\\0\\4\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0"... , 78
4, 64) = 784
12:15:52 mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f7517ac9000
12:15:52 mprotect(0x7f7517af1000, 2023424, PROT_NONE) = 0
12:15:52 mmap(0x7f7517af1000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENY
WRITE, 3, 0x28000) = 0x7f7517af1000
12:15:52 mmap(0x7f7517c86000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x1bd000) = 0x7f7517c86000
12:15:52 mmap(0x7f7517cdf000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0x215000) = 0x7f7517cdf000
12:15:52 mmap(0x7f7517ce5000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONY
MOUS, -1, 0) = 0x7f7517ce5000
12:15:52 close(3) = 0
12:15:52 mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -
1, 0) = 0x7f7517ac6000
12:15:52 arch_prctl(ARCH_SET_FS, 0x7f7517ac6740) = 0
12:15:52 set_tid_address(0x7f7517ac6a10) = 8932
12:15:52 set_robust_list(0x7f7517ac6a20, 24) = 0
12:15:52 rseq(0x7f7517ac70e0, 0x20, 0, 0x53053053) = 0
12:15:52 mprotect(0x7f7517cdf000, 16384, PROT_READ) = 0
12:15:52 mprotect(0x55df13bdd000, 4096, PROT_READ) = 0
12:15:52 mprotect(0x7f7517d30000, 8192, PROT_READ) = 0
12:15:52 prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY})
= 0
12:15:52 munmap(0x7f7517cf2000, 12460) = 0

```

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12:15:52 getrandom("\x20\x33\xc8\x61\x2a\xd0\xb7\x64", 8, GRND_NONBLOCK) = 8
12:15:52 brk(NULL) = 0x55df156db000
12:15:52 brk(0x55df156fc000) = 0x55df156fc000
Подгруз библиотеки 1
12:15:52 openat(AT_FDCWD, "./libimpl1.so", O_RDONLY|O_CLOEXEC) = 3
12:15:52 read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832)
= 832
12:15:52 newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=15760, ...}, AT_EMPTY_PATH) = 0
12:15:52 getcwd("/workspaces/Operating_Systems/lab4/build/bin", 128) = 45
12:15:52 mmap(NULL, 16448, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f7517ac1000
12:15:52 mmap(0x7f7517ac2000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x1000) = 0x7f7517ac2000
12:15:52 mmap(0x7f7517ac3000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2
000) = 0x7f7517ac3000
12:15:52 mmap(0x7f7517ac4000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x2000) = 0x7f7517ac4000
12:15:52 close(3) = 0
12:15:52 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
12:15:52 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
12:15:52 mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f7517cf2000
12:15:52 close(3) = 0
Подгруз математики
12:15:52 openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
12:15:52 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832)
= 832
12:15:52 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) = 0
12:15:52 mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f75179da000
12:15:52 mmap(0x7f75179e8000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0xe000) = 0x7f75179e8000
12:15:52 mmap(0x7f7517a64000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x8a000) = 0x7f7517a64000
12:15:52 mmap(0x7f7517abf000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0xe4000) = 0x7f7517abf000
12:15:52 close(3) = 0
12:15:52 mprotect(0x7f7517abf000, 4096, PROT_READ) = 0
12:15:52 mprotect(0x7f7517ac4000, 4096, PROT_READ) = 0
12:15:52 munmap(0x7f7517cf2000, 12460) = 0
12:15:52 newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY
_PATH) = 0
12:15:52 write(1, "=== \320\237\321\200\320\276\320\263\321\200\320\260\320\274\320\274\32
0\260 \321\201 \320\264\320\270\320\275"..., 93=== Программа с динамической загрузкой библи
отек ===
) = 93
12:15:52 write(1, "\320\242\320\265\320\272\321\203\321\211\320\260\321\217 \320\261\320\2
70\320\261\320\273\320\270\320\276\321\202\320\265\320"..., 51Текущая библиотека: ./libimp
l1.so
) = 51
12:15:52 write(1, "\320\232\320\276\320\274\320\260\320\275\320\264\321\213:\n", 16Команды
:
) = 16
12:15:52 write(1, " 0 - \320\277\320\265\321\200\320\265\320\272\320\273\321\216\321\207\3
20\270\321\202\321\214 \320\261\320"..., 50 0 - переключить библиотеку
) = 50
12:15:52 write(1, " 1 \321\207\320\270\321\201\320\273\320\276 - \320\277\320\265\321\200
\320\265\320\262\320\276\320\264 "..., 57 1 число - перевод числа (translation)
) = 57
12:15:52 write(1, " 2 A B e - \320\270\320\275\321\202\320\265\320\263\321\200\320\260\32
0\273 sin"..., 50 2 A B e - интеграл sin(x) (SinIntegral)
) = 50
12:15:52 write(1, " 9 - \320\262\321\213\321\205\320\276\320\264\n", 17 9 - выход
) = 17
12:15:52 write(1, "\n", 1

```

```

) = 1
12:15:52 newfstatat(0, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}, AT_EMPTY_PATH) = 0
12:15:52 write(1, "> ", 2) = 2
12:15:52 read(0, 1 7
"1 7\n", 1024) = 4
12:15:57 write(1, "translation(7) = 111\n", 21translation(7) = 111
) = 21
12:15:57 write(1, "> ", 2) = 2
Поступил 0
12:15:57 read(0, 0
"0\n", 1024) = 2
12:16:07 munmap(0x7f7517ac1000, 16448) = 0
12:16:07 munmap(0x7f75179da000, 942344) = 0
Переключаем библиотеку
12:16:07 openat(AT_FDCWD, "./libimpl2.so", O_RDONLY|O_CLOEXEC) = 3
12:16:07 read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832)
= 832
12:16:07 newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=15760, ...}, AT_EMPTY_PATH) = 0
12:16:07 getcwd("/workspaces/Operating_Systems/lab4/build/bin", 128) = 45
12:16:07 mmap(NULL, 16448, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f7517ac1000
12:16:07 mmap(0x7f7517ac2000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1000) = 0x7f7517ac2000
12:16:07 mmap(0x7f7517ac3000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7f7517ac3000
12:16:07 mmap(0x7f7517ac4000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7f7517ac4000
12:16:07 close(3) = 0
12:16:07 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
12:16:07 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
12:16:07 mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f7517cf2000
12:16:07 close(3) = 0
12:16:07 openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
12:16:07 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"..., 832)
= 832
12:16:07 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) = 0
12:16:07 mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f75179da000
12:16:07 mmap(0x7f75179e8000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe000) = 0x7f75179e8000
12:16:07 mmap(0x7f7517a64000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x8a000) = 0x7f7517a64000
12:16:07 mmap(0x7f7517abf000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe4000) = 0x7f7517abf000
12:16:07 close(3) = 0
12:16:07 mprotect(0x7f7517abf000, 4096, PROT_READ) = 0
12:16:07 mprotect(0x7f7517ac4000, 4096, PROT_READ) = 0
12:16:07 munmap(0x7f7517cf2000, 12460) = 0
12:16:07 write(1, "\320\237\320\265\321\200\320\265\320\272\320\273\321\216\321\207\320\265\320\275\320\276 \320\275\320\260 \320\261\320\270"..., 64Переключено на библиотеку: ./libimpl2.so
) = 64
12:16:07 write(1, "> ", 2) = 2
12:16:07 read(0, 1 7
"1 7\n", 1024) = 4
12:16:11 write(1, "translation(7) = 21\n", 20translation(7) = 21
) = 20
12:16:11 write(1, "> ", 2) = 2
12:16:11 read(0, 2 0 1 0.1
"2 0 1 0.1\n", 1024) = 10
12:16:41 write(1, "[\320\221\320\270\320\261\320\273\320\270\320\276\321\202\320\265\320\272\320\260 2] \320\234\320\265\321\202\320"..., 53[Библиотека 2] Метод ТРАПЕЦИЙ
) = 53

```

```

12:16:41 write(1, "SinIntegral(0.00, 1.00, 0.1000) "..., 43SinIntegral(0.00, 1.00, 0.1000)
= 0.459315
) = 43
12:16:41 write(1, "> ", 2> ) = 2
Поступил 0
12:16:41 read(0, 0
"0\n", 1024) = 2
Высвобождаем 2 библиотеку
12:16:45 munmap(0x7f7517ac1000, 16448) = 0
12:16:45 munmap(0x7f75179da000, 942344) = 0
Загружаем 1
12:16:45 openat(AT_FDCWD, "./libimpl1.so", O_RDONLY|O_CLOEXEC) = 3
12:16:45 read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0"..., 832)
= 832
12:16:45 newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=15760, ...}, AT_EMPTY_PATH) = 0
12:16:45 getcwd("/workspaces/Operating_Systems/lab4/build/bin", 128) = 45
12:16:45 mmap(NULL, 16448, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f7517ac1000
12:16:45 mmap(0x7f7517ac2000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x1000) = 0x7f7517ac2000
12:16:45 mmap(0x7f7517ac3000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2
000) = 0x7f7517ac3000
12:16:45 mmap(0x7f7517ac4000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0x2000) = 0x7f7517ac4000
12:16:45 close(3) = 0
12:16:45 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
12:16:45 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=12460, ...}, AT_EMPTY_PATH) = 0
12:16:45 mmap(NULL, 12460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f7517cf2000
12:16:45 close(3) = 0
12:16:45 openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
12:16:45 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0"..., 832)
= 832
12:16:45 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=940560, ...}, AT_EMPTY_PATH) = 0
12:16:45 mmap(NULL, 942344, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f75179da000
12:16:45 mmap(0x7f75179e8000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0xe000) = 0x7f75179e8000
12:16:45 mmap(0x7f7517a64000, 372736, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x8a000) = 0x7f7517a64000
12:16:45 mmap(0x7f7517abf000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWR
ITE, 3, 0xe4000) = 0x7f7517abf000
12:16:45 close(3) = 0
12:16:45 mprotect(0x7f7517abf000, 4096, PROT_READ) = 0
12:16:45 mprotect(0x7f7517ac4000, 4096, PROT_READ) = 0
12:16:45 munmap(0x7f7517cf2000, 12460) = 0
12:16:45 write(1, "\320\237\320\265\321\200\320\265\320\272\320\273\321\216\321\207\320\26
5\320\275\320\276 \320\275\320\260 \320\261\320\270"..., 64Переключено на библиотеку: ./li
bimpl1.so
) = 64
12:16:45 write(1, "> ", 2> ) = 2
12:16:45 read(0, 2 0 2 0.1
"2 0 2 0.1\n", 1024) = 10
12:16:51 write(1, "[\320\221\320\270\320\261\320\273\320\270\320\276\321\202\320\265\320\2
72\320\260 1] \320\234\320\265\321\202\320"..., 67[Библиотека 1] Метод ПРЯМОУГОЛЬНИКОВ
) = 67
12:16:51 write(1, "SinIntegral(0.00, 2.00, 0.1000) "..., 43SinIntegral(0.00, 2.00, 0.1000)
= 1.416737
) = 43
12:16:51 write(1, "> ", 2> ) = 2
12:16:51 read(0, 9
"9\n", 1024) = 2
12:16:54 write(1, "\320\222\321\213\321\205\320\276\320\264 \320\270\320\267 \320\277\321\
200\320\276\320\263\321\200\320\260\320\274\320\274"..., 35Выход из программы
) = 35

```

Освобождаем библиотеку

```
12:16:54 munmap(0x7f7517ac1000, 16448) = 0
12:16:54 munmap(0x7f75179da000, 942344) = 0
12:16:54 exit_group(0)                = ?
12:16:54 +++ exited with 0 +++
```

Вывод

В ходе выполнения лабораторной работы №5 был проведён детальный анализ системных вызовов программ, разработанных в предыдущих четырёх лабораторных работах, с помощью утилиты strace. Это позволило наглядно увидеть, как высокоуровневые операции на языке Си (создание процессов, работа с каналами, разделяемой памятью, потоками и динамическими библиотеками) транслируются в низкоуровневые запросы к ядру операционной системы, такие как clone, pipe2, mmap, openat и munmap. Анализ подтвердил корректность реализации межпроцессного взаимодействия и управления ресурсами в каждой из рассмотренных программ, а также выявил характерные паттерны использования системных вызовов для разных механизмов коммуникации (pipes, shared memory) и многозадачности (процессы vs потоки).